

Expert System And Machine Vision

Chapter 6

- An Expert System is a branch of Artificial Intelligence designed to simulate the decision-making ability of a human expert.
- Works like a human specialist like a doctor, engineer, lawyer, etc.
- Uses stored knowledge and logical rules to solve problems.
- Gives consistent and quick decisions.
- Mostly used in diagnosis, troubleshooting and planning.

Architecture of an Expert System

- **User Interface (UI)**
- Allows communication between the user and the expert system.
- Takes user input and displays recommendations.
- **Example:**
Doctor enters patient symptoms.

Knowledge Base

- The **Knowledge Base** stores all expert knowledge.
- It contains:
 - Facts
 - Rules
 - Relationships
 - Procedures
- It is the "brain" of the expert system.
- **Example Rule**
 - IF
 - Fever = Yes
 - Cough = Yes
 - THEN
 - Disease = Flu

Inference Engine

- The **Inference Engine** is the reasoning part.
- Matching user input with rules in the knowledge base
- Applying logical reasoning to derive new facts
- Repeating the reasoning process until a conclusion is found
- It acts like the thinking process of a human expert.
- **Methods Used**
 - Forward Chaining
 - Backward Chaining

Explanation Module

- One of the key advantages of expert systems over many modern AI models is their ability to explain their reasoning. The explanation system helps users understand:
 - Why the system asked a particular question
 - How it reached a specific conclusion
 - Which rules were applied
 - What facts influenced the result

Stages of Expert System Development

- **Stage 1: Problem Identification**
- Define the problem.
- Decide whether an expert system is suitable.
- Determine objectives.
- **Example:** Develop a system to diagnose plant diseases.
- **Stage 2: Knowledge Acquisition**
- Collect knowledge from:
 - Human experts
 - Books
 - Journals
 - Databases
 - Previous cases
- This is often the most time-consuming stage.

Stage 3: Knowledge Representation

- Convert collected knowledge into a form the computer can use.
- Methods include:
- IF–THEN Rules
- Frames
- Semantic Networks
- Logic
- Example:
- IF Temperature > 100°F
- THEN Fever = Yes

Stage 4: System Design

- Design the architecture.
- Create:
 - Knowledge Base
 - Inference Engine
 - User Interface
 - Working Memory
 - Explanation Facility

Stage 5: Implementation (Coding)

- Develop the expert system using programming languages or expert system shells.
- Examples:
 - Prolog
 - Lisp
 - Python
 - CLIPS

Stage 6: Testing and Validation

- Check whether:
- Rules are correct.
- Conclusions are accurate.
- The system behaves like a human expert.

Stage 7: Deployment

- Install the expert system for actual users.
- Users begin solving real problems with it.

Stage 8: Maintenance and Updating

- Continuously update:
- New rules
- New knowledge
- Bug fixes
- Performance improvements
- This stage keeps the system useful as knowledge evolves.

Concept of Machine Vision

- Machine Vision is a branch of Artificial Intelligence (AI) and computer vision that enables a computer or machine to capture, process, analyze, and understand images so it can make decisions automatically without human intervention.
- Machine Vision is the technology that enables computers or machines to acquire images, process them, extract useful information, and make decisions without human intervention.

Steps of Machine Vision

Image Acquisition

- A camera captures an image of the object.
- Proper lighting is essential for obtaining a high-quality image.

Image Pre-processing

- The captured image is improved before analysis.
- Common operations include:
 - Removing noise
 - Increasing brightness
 - Improving contrast
 - Filtering
 - Resizing

Image Segmentation

- The system separates the important object from the background.
- For example:
 - Separate an apple from the table.
 - Separate a coin from the background.
- This helps the computer focus only on the required object.

Feature Extraction

- The computer identifies important characteristics of the object.
- Examples of features:
 - Shape
 - Size
 - Color
 - Edges
 - Texture
 - Corners
- These features help identify or classify the object

Object Recognition / Classification

- The extracted features are compared with stored patterns.
- The system determines:
 - What the object is
 - Whether it is defective
 - Which category it belongs to
- Example:
- Good bottle
- Broken bottle

Decision Making

- The machine decides what action should be taken.
- Examples:
 - Accept the product
 - Reject the defective product
 - Count objects
 - Measure dimensions
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Tutorial

- Application of machine
- Hard vs. Strong AI, Soft vs. Weak AI